

QUICKLOAN MOBILE

Ethical Data Governance Review

Independent Data Governance Consultant Report

Confidential | May 2026

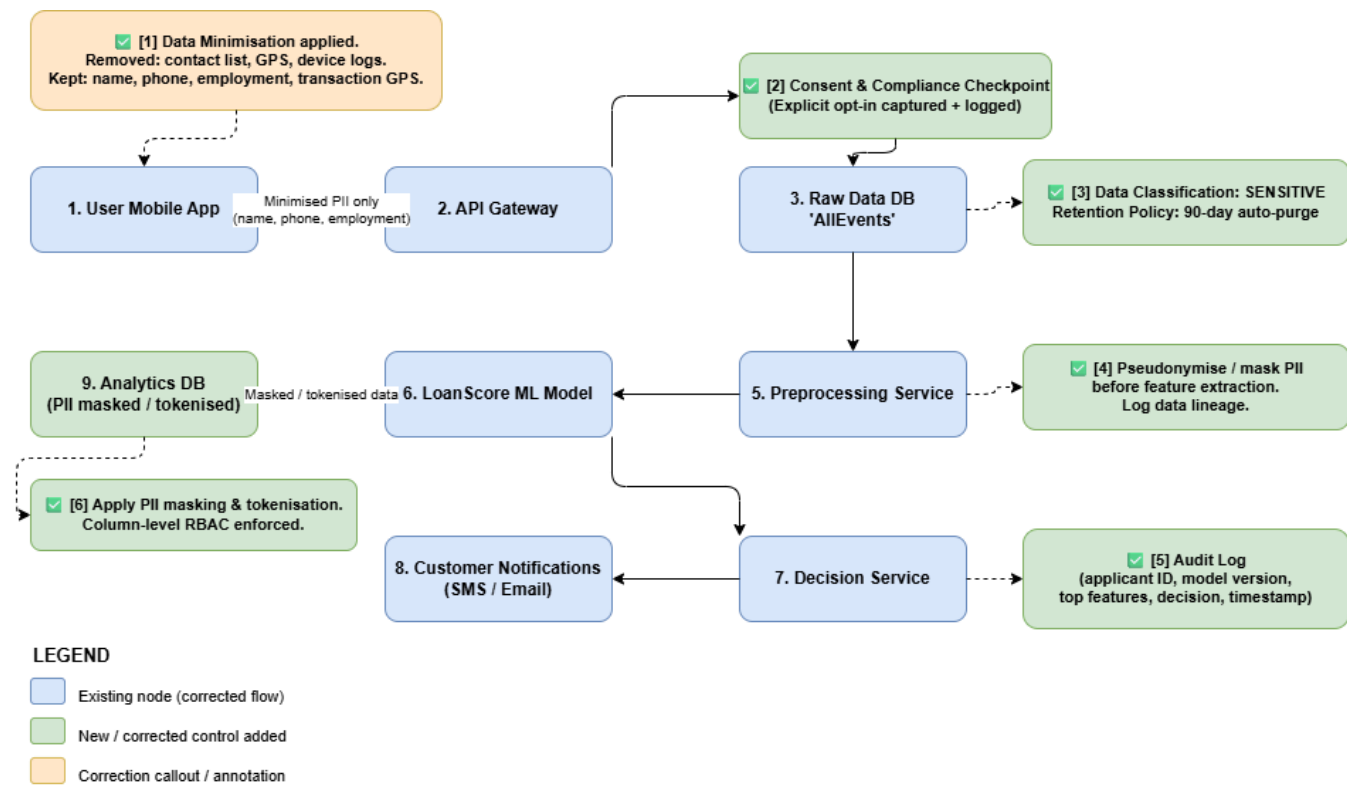
Deliverable 1: Governance Review Card

This card captures three distinct risk findings identified during the review of QuickLoan Mobile’s data pipeline, together with a reporting recommendation to ensure ongoing ethical governance transparency.

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
1. DATA QUALITY RISK			
Specific Quality Issue	Incomplete & Inconsistent Customer Records Many customer profiles arrive from the mobile app with missing fields (e.g., no employment data, blank date-of-birth) and inconsistent formats (phone numbers without country code, dates mixed as DD/MM/YYYY and YYYY-MM-DD).	Corrupted training data fed into the LoanScore ML model, producing unreliable credit scores. Loan decisions made on bad data expose the company to financial loss and discriminatory outcomes that cannot be easily audited.	Implement schema validation & data quality gates at the API Gateway (Step 2). Reject or quarantine records failing format checks before they enter the Raw Data DB. Apply standardised data-entry rules in the mobile app (input masks, mandatory fields). Schedule monthly data-quality audits using a completeness KPI dashboard.
2. LEGAL & COMPLIANCE RISK			
Data Classification	SENSITIVE The pipeline collects full name, phone, GPS location, complete contact list, device logs, and financial history — all constituting Special Category / Sensitive PII under Ghana’s Data Protection Act (DPA), Act 843.	Breach of Act 843 attracts fines, reputational damage, and potential criminal liability. The absence of a consent mechanism and the collection of an entire contact list (third-party PII) are clear violations of the Necessity and Proportionality principles.	Immediate actions: - Insert a granular, layered consent screen before data collection starts (explicit opt-in per data type). - Apply Data Minimisation: remove contact list collection entirely; restrict GPS to transaction-time only. - Appoint a Data Protection Officer (DPO) and register with the Data

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			Protection Commission. Implement a 90-day retention limit on Raw Event data with automated purge jobs.
3. BIAS & FAIRNESS RISK			
Source of Bias	Proxy-variable bias in the LoanScore ML Model (Step 6) Features such as GPS home location (postal code), contact-list composition, and device type act as proxies for protected attributes (ethnicity, gender, socioeconomic class). The model is trained on historically biased approval data, encoding past discriminatory patterns.	Systematic under-approval or predatory pricing for already-marginalised communities in Accra, Greater Accra, and Northern regions. This violates fairness obligations under Act 843 and exposes QuickLoan to regulatory enforcement and class-action risk.	Proposed monitoring steps: - Remove or audit proxy features (GPS, contact list) before model training. - Run a monthly Disparate Impact Audit: calculate approval rate by gender, region, and age cohort. - Enforce a Disparate Impact Ratio (DIR) threshold ≥ 0.80 (80% Rule); trigger model review if breached. - Log every model decision with feature weights to enable post-hoc explainability audits.
4. STORYTELLING / REPORTING RECOMMENDATION			
Metric to Monitor	Approval Rate Disparity Index (ARDI) Definition: The ratio of the loan approval rate for the lowest-approval demographic cohort to the approval rate for the highest-approval cohort, calculated monthly per gender, age group, and region. Formula: $ARDI = \frac{\text{Min}(\text{group approval rate})}{\text{Max}(\text{group approval rate})}$. Healthy threshold: $ARDI \geq 0.80$.	Visualization Type: Grouped Bar Chart (monthly snapshots) with a horizontal reference line at 0.80, broken down by demographic segment. Overlay a trend line to show improvement or deterioration across quarters.	Why It Matters: The ARDI translates algorithmic fairness into a single, boardroom-ready number that non-technical stakeholders can monitor, making systemic discrimination visible and actionable before it escalates into a regulatory enforcement event.

Deliverable 2: Corrected Data Flow — Annotated Corrections



3: Summary of Review Process

Applying Data Lifecycle and Classification Principles to Surface Risk

This review was structured around the full data lifecycle — Collection, Storage, Processing, Use, and Retention — treating each stage as both a governance checkpoint and a potential failure point. The first step was to map QuickLoan's pipeline against this lifecycle to identify where data was being collected, how it moved, and crucially, where controls were absent.

At the Collection stage (Step 1), Data Minimisation principles immediately flagged the contact-list harvest and persistent GPS tracking as proportionality violations. These features are neither necessary nor relevant to a micro-loan scoring decision, making their collection indefensible under Ghana's Data Protection Act, Act 843. Applying a Classification lens to the data confirmed that fields such as national ID, location, and financial history constitute Sensitive PII, triggering the highest tier of governance obligations: explicit consent, encryption at rest and in transit, strict access controls, and defined retention limits.

At the Storage stage (Step 3 and Step 9), the absence of any data-classification tags or retention schedules meant the Raw Data DB and Analytics DB were effectively ungoverned. Without classification, there is no logical basis for access control or audit prioritisation. Introducing 90-day retention policies and pseudonymisation before analytics ingestion directly addresses both the legal storage-limitation requirement and the practical risk of a broad-surface data breach.

At the Processing and Decision stages (Steps 5 and 7), the lifecycle lens revealed the most insidious risk: the LoanScore ML model was being trained on unaudited, potentially proxy-biased features without any decision logging. A model that cannot be inspected cannot be held accountable.

The proposed reporting metric — the Approval Rate Disparity Index (ARDI) — directly addresses this accountability gap. By computing the ratio between the lowest and highest group-level approval rates monthly, the ARDI translates abstract fairness principles into a single, auditable number. A threshold of 0.80 is widely recognised in fairness literature and is operationally testable. Visualised as a grouped bar chart with a reference line, the metric makes bias trends legible to non-technical stakeholders: board members, regulators, and customers alike. Ethical governance is only meaningful when it is observable, and the ARDI ensures that algorithmic decisions remain subject to continuous, transparent scrutiny rather than being treated as a black box.